

scanFiles

June 19, 2025

```
[2]: import os
import re
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
[147]: def count_words_in_go_files(folder_path, words_dict):
    word_search_counts = {word: 0 for word in words_dict.keys()}
    word_markup_counts = words_dict.copy()

    pattern = re.compile(r'\b(' + '|'.join(map(re.escape, words_dict.keys())) +
    ↪r')\b')

    for root, dirs, files in os.walk(folder_path):
        for file in files:
            if file.endswith('.go'):
                file_path = os.path.join(root, file)
                try:
                    with open(file_path, 'r', encoding='utf-8') as f:
                        content = f.read()
                        matches = pattern.findall(content)
                        for word in matches:
                            word_search_counts[word] += 1
                except (UnicodeDecodeError, PermissionError) as e:
                    print(f"                {file_path}: {e}")

    # DataFrame
    df = pd.DataFrame({
        'Word': list(word_search_counts.keys()),
        'Count Search': list(word_search_counts.values()),
        'Count Markup': [word_markup_counts[word] for word in
    ↪word_search_counts.keys()]
    })

    # Ratio
    df['Ratio'] = df['Count Search'] / df['Count Markup']
```

```

median = df['Ratio'].median()
mad = (df['Ratio'] - median).abs().median()
threshold = 3
df['IsOutlier_MAD'] = (df['Ratio'] - median).abs() > (threshold * mad)
df['ratio<30'] = df['Ratio']<30

#           Ratio (           )
df_sorted = df.sort_values(by='Ratio', ascending=False)
df_sorted.reset_index(drop=True, inplace=True)

return df_sorted

```

```

[148]: global_array = np.array([])
       one_array = np.array([])

```

```

[149]: def analyze(folder, words):
       result_df = count_words_in_go_files(folder, words)

       mean_row = pd.DataFrame({
           'Word': ['MEAN'],
           'Count Search': [result_df['Count Search'].mean()],
           'Count Markup': [result_df['Count Markup'].mean()],
           'Ratio': [result_df['Ratio'].mean()]
       })

       global global_array
       global_array = np.append(global_array, result_df['Ratio'].values)

       display(result_df)
       display(mean_row)
       s1 = iterate_over_column(result_df[result_df['ratio<30']]["Count Search"],_
       ↪dt1)
       s2 = iterate_over_column(result_df[result_df['ratio<30']]["Count Markup"],_
       ↪dt2)
       print(s1/s2)

```

```

[161]: #           (           )
       dt1 = 4000
       #           (           ),
       dt2 = 3000
       #
       dt = 500
       #
       dd = 100

```

```
[162]: def iterate_over_column(column: pd.Series, dtX):
    s = 0
    for val in column:
        if val == 1:
            s += val * 0 + dt + dd
        else:
            s += val * dtX + dt + dd
    return s
```

```
[163]: folder = 'e:\\phd\\test_repos_graphql2\\boost\\'
words = {
    "DealUpdate": 1,
    "StorageAskUpdate": 1,
    "FundsMoveToEscrow": 1,
    "DealFailPaused": 1,
    "DealRetryPaused": 1,
    "DealCancel": 1,
    "Mpool": 1,
    "FundsLogs": 1,
    "ProposalLogs": 1,
    "LegacyDeals": 1,
    "Deals": 1,
    "LegacyDeal": 1,
    "Deal": 2,
    "DealNew": 1,
    "DealsCount": 1,
    "StorageAsk": 1,
    "DealPublishNow": 1,
    "DealPublish": 1,
    "Epoch": 1,
    "Funds": 1,
    "ProposalLogsCount": 1,
    "SealingPipeline": 1,
    "Storage": 1,
    "Transfers": 1,
}
analyze(folder, words)
```

	Word	Count	Search	Count	Markup	Ratio	IsOutlier_MAD	\
0	Deals		56		1	56.0	True	
1	StorageAsk		32		1	32.0	True	
2	Storage		26		1	26.0	True	
3	Deal		16		2	8.0	True	
4	Epoch		7		1	7.0	True	
5	ProposalLogs		5		1	5.0	True	
6	FundsLogs		4		1	4.0	True	
7	Transfers		4		1	4.0	True	
8	DealNew		2		1	2.0	False	

9	DealPublish	2	1	2.0	False
10	Funds	2	1	2.0	False
11	SealingPipeline	2	1	2.0	False
12	Mpool	1	1	1.0	False
13	LegacyDeals	1	1	1.0	False
14	DealCancel	1	1	1.0	False
15	LegacyDeal	1	1	1.0	False
16	StorageAskUpdate	1	1	1.0	False
17	DealsCount	1	1	1.0	False
18	DealRetryPaused	1	1	1.0	False
19	DealPublishNow	1	1	1.0	False
20	DealFailPaused	1	1	1.0	False
21	ProposalLogsCount	1	1	1.0	False
22	FundsMoveToEscrow	1	1	1.0	False
23	DealUpdate	1	1	1.0	False

```

ratio<30
0      False
1      False
2      True
3      True
4      True
5      True
6      True
7      True
8      True
9      True
10     True
11     True
12     True
13     True
14     True
15     True
16     True
17     True
18     True
19     True
20     True
21     True
22     True
23     True

```

	Word	Count	Search	Count	Markup	Ratio
0	MEAN		7.083333		1.041667	6.75

15.270833333333334

```
[164]: folder = 'e:\\phd\\test_repos_graphql2\\circuit\\'
words = {
    "Secrets": 2,
    "Services": 2,
    "Releases": 2,
    "EnvsDeployedIn": 2,
    "Features": 2,
    "GetGitCommits": 1,
    "BookmarkProject": 1,
    "UpdateProjectEnvironments": 1,
    "UpdateUserPermissions": 1,
    "DeleteProjectExtension": 1,
    "UpdateProjectExtension": 1,
    "CreateProjectExtension": 1,
    "DeleteExtension": 1,
    "UpdateExtension": 1,
    "CreateExtension": 1,
    "ImportSecrets": 1,
    "DeleteSecret": 1,
    "UpdateSecret": 1,
    "CreateSecret": 1,
    "DeleteEnvironment": 1,
    "UpdateEnvironment": 1,
    "CreateEnvironment": 1,
    "DeleteServiceSpec": 1,
    "UpdateServiceSpec": 1,
    "CreateServiceSpec": 1,
    "ImportServices": 1,
    "DeleteService": 1,
    "UpdateService": 1,
    "CreateService": 1,
    "StopRelease": 1,
    "CreateRelease": 1,
    "UpdateProject": 1,
    "CreateProject": 1,
    "Permissions": 1,
    "ReleaseExtensions": 1,
    "ProjectExtensions": 1,
    "Extensions": 1,
    "Secret": 1,
    "ExportSecrets": 1,
    "Environments": 1,
    "ServiceSpecs": 1,
    "Users": 1,
    "User": 1,
    "Bookmarked": 1,
}
```

```
analyze(folder, words)
```

	Word	Count	Search	Count	Markup	Ratio	\
0	User		2577		1	2577.0	
1	Secret		660		1	660.0	
2	Extensions		379		1	379.0	
3	Permissions		371		1	371.0	
4	Services		541		2	270.5	
5	CreateProject		180		1	180.0	
6	CreateEnvironment		164		1	164.0	
7	Secrets		299		2	149.5	
8	CreateService		143		1	143.0	
9	Users		122		1	122.0	
10	CreateSecret		99		1	99.0	
11	UpdateService		82		1	82.0	
12	UpdateProject		66		1	66.0	
13	CreateExtension		61		1	61.0	
14	CreateServiceSpec		55		1	55.0	
15	UpdateSecret		54		1	54.0	
16	CreateProjectExtension		54		1	54.0	
17	UpdateEnvironment		41		1	41.0	
18	DeleteService		39		1	39.0	
19	CreateRelease		37		1	37.0	
20	Environments		33		1	33.0	
21	Features		62		2	31.0	
22	DeleteSecret		27		1	27.0	
23	ProjectExtensions		26		1	26.0	
24	DeleteEnvironment		21		1	21.0	
25	Bookmarked		17		1	17.0	
26	ImportServices		12		1	12.0	
27	ReleaseExtensions		12		1	12.0	
28	StopRelease		12		1	12.0	
29	ImportSecrets		10		1	10.0	
30	UpdateProjectEnvironments		10		1	10.0	
31	Releases		19		2	9.5	
32	UpdateUserPermissions		9		1	9.0	
33	DeleteServiceSpec		9		1	9.0	
34	DeleteProjectExtension		8		1	8.0	
35	UpdateProjectExtension		8		1	8.0	
36	UpdateServiceSpec		8		1	8.0	
37	BookmarkProject		8		1	8.0	
38	DeleteExtension		7		1	7.0	
39	ExportSecrets		7		1	7.0	
40	UpdateExtension		7		1	7.0	
41	ServiceSpecs		4		1	4.0	
42	EnvsDeployedIn		8		2	4.0	
43	GetGitCommits		3		1	3.0	

	IsOutlier_MAD	ratio<30
0	True	False
1	True	False
2	True	False
3	True	False
4	True	False
5	True	False
6	True	False
7	True	False
8	True	False
9	True	False
10	True	False
11	False	False
12	False	False
13	False	False
14	False	False
15	False	False
16	False	False
17	False	False
18	False	False
19	False	False
20	False	False
21	False	False
22	False	True
23	False	True
24	False	True
25	False	True
26	False	True
27	False	True
28	False	True
29	False	True
30	False	True
31	False	True
32	False	True
33	False	True
34	False	True
35	False	True
36	False	True
37	False	True
38	False	True
39	False	True
40	False	True
41	False	True
42	False	True
43	False	True

	Word	Count	Search	Count	Markup	Ratio
0	MEAN	144.795455		1.113636	134.238636	

40.523809523809526

```
[165]: folder = 'e:\\phd\\test_repos_graphql2\\core-geth\\'
words = {
    "SendRawTransaction": 2,
    "Logs": 2,
    "Transaction": 2,
    "Blocks": 1,
    "Block": 1,
    "EstimateGas": 1,
    "Call": 1,
    "Account": 1,
    "EstimateGas": 1,
    "Call": 1,
    "Account": 1,
    "OmmerAt": 1,
    "Miner": 1,
    "CreatedContract": 1,
    "Account": 1,
    "Storage": 1,
    "Status": 1,
    "Nonce": 1,
    "Hash": 1,
    "TransactionCount": 1,
    "ChainID": 1,
    "MaxPriorityFeePerGas": 1,
    "GasPrice": 1,
    "Transactions": 1,
    "ExcessBlobGas": 1,
    "BlobGasUsed": 1,
    "Withdrawals": 1,
    "WithdrawalsRoot": 1,
    "Raw": 1,
    "RawHeader": 1,
    "TotalDifficulty": 1,
    "LogsBloom": 1,
    "ExtraData": 1,
    "Ommers": 1,
    "OmmerCount": 1,
    "OmmerHash": 1,
    "ReceiptsRoot": 1,
    "StateRoot": 1,
    "TransactionsRoot": 1,
    "MixHash": 1,
    "Timestamp": 1,
    "Difficulty": 1,
    "Parent": 1,
```



```

    "NextBaseFeePerGas": 1,
    "BaseFeePerGas": 1,
    "GasUsed": 1,
    "GasLimit": 1,
    "Number": 1,
    "YParity": 1,
    "AccessList": 1,
    "Type": 1,
    "BlobGasPrice": 1,
    "CumulativeGasUsed": 1,
    "Index": 1,
    "Value": 1,
    "EffectiveTip": 1,
    "BlobVersionedHashes": 1,
    "MaxFeePerBlobGas": 1,
    "MaxFeePerGas": 1,
    "EffectiveGasPrice": 1,
    "Gas": 1,
    "InputData": 1
}
analyze(folder, words)

```

	Word	Count	Search	Count	Markup	Ratio	IsOutlier_MAD	\
0	Hash		6364		1	6364.0	True	
1	Number		1446		1	1446.0	True	
2	Value		1096		1	1096.0	True	
3	Type		1049		1	1049.0	True	
4	Block		770		1	770.0	True	
5	Nonce		712		1	712.0	True	
6	Account		442		1	442.0	True	
7	Transaction		852		2	426.0	True	
8	Gas		420		1	420.0	True	
9	Difficulty		387		1	387.0	True	
10	Transactions		345		1	345.0	True	
11	GasLimit		341		1	341.0	True	
12	GasPrice		319		1	319.0	True	
13	Storage		253		1	253.0	False	
14	ChainID		246		1	246.0	False	
15	GasUsed		219		1	219.0	False	
16	Timestamp		188		1	188.0	False	
17	AccessList		182		1	182.0	False	
18	Call		169		1	169.0	False	
19	Status		168		1	168.0	False	
20	Index		156		1	156.0	False	
21	Withdrawals		148		1	148.0	False	
22	ExcessBlobGas		146		1	146.0	False	
23	BlobGasUsed		131		1	131.0	False	
24	Miner		105		1	105.0	False	

25	Logs	164	2	82.0	False
26	MaxFeePerGas	82	1	82.0	False
27	ExtraData	73	1	73.0	False
28	MaxPriorityFeePerGas	72	1	72.0	False
29	Parent	60	1	60.0	False
30	Blocks	59	1	59.0	False
31	CumulativeGasUsed	51	1	51.0	False
32	StateRoot	51	1	51.0	False
33	Raw	47	1	47.0	False
34	BaseFeePerGas	32	1	32.0	False
35	LogsBloom	23	1	23.0	False
36	EffectiveGasPrice	23	1	23.0	False
37	TotalDifficulty	18	1	18.0	False
38	BlobGasPrice	17	1	17.0	False
39	EstimateGas	17	1	17.0	False
40	BlobVersionedHashes	17	1	17.0	False
41	Ommers	17	1	17.0	False
42	MixHash	16	1	16.0	False
43	ReceiptsRoot	16	1	16.0	False
44	WithdrawalsRoot	14	1	14.0	False
45	YParity	12	1	12.0	False
46	OmmmerHash	12	1	12.0	False
47	TransactionCount	9	1	9.0	False
48	MaxFeePerBlobGas	8	1	8.0	False
49	TransactionsRoot	4	1	4.0	False
50	EffectiveTip	2	1	2.0	False
51	SendRawTransaction	4	2	2.0	False
52	NextBaseFeePerGas	2	1	2.0	False
53	OmmmerCount	2	1	2.0	False
54	RawHeader	2	1	2.0	False
55	CreatedContract	2	1	2.0	False
56	OmmmerAt	2	1	2.0	False
57	InputData	2	1	2.0	False

ratio<30	
0	False
1	False
2	False
3	False
4	False
5	False
6	False
7	False
8	False
9	False
10	False
11	False
12	False

13	False
14	False
15	False
16	False
17	False
18	False
19	False
20	False
21	False
22	False
23	False
24	False
25	False
26	False
27	False
28	False
29	False
30	False
31	False
32	False
33	False
34	False
35	True
36	True
37	True
38	True
39	True
40	True
41	True
42	True
43	True
44	True
45	True
46	True
47	True
48	True
49	True
50	True
51	True
52	True
53	True
54	True
55	True
56	True
57	True

	Word	Count	Search	Count	Markup	Ratio
0	MEAN	303.206897		1.051724	294.413793	

49.38383838383838

```
[166]: folder = 'e:\\phd\\test_repos_graphql2\\exo\\'
words = {
    "Components": 2,
    "FindManifest": 1,
    "WatchJob": 1,
    "StackByID": 1,
    "DestroyComponents": 1,
    "UpdateComponent": 2,
    "DestroyStack": 1,
    "FinishTask": 1,
    "AcquireTask": 1,
    "DisposeResource": 1,
    "UpdateResource": 2,
    "RestartWorkspaceComponents": 1,
    "CreateWorkspace": 2,
    "CreateProject": 1,
    "ResourceByID": 1,
    "MakeManifest": 1,
    "FindEntity": 1,
    "Stack": 1,
    "Manifest": 1,
    "BuildWorkspace": 1,
    "DestroyWorkspace": 1,
    "TaskByID": 1,
    "DisplayName": 1,
    "ForgetResource": 1,
    "ReconcileComponent": 1,
    "InitializeResource": 1,
    "RefreshResource": 1,
    "RefreshCluster": 1,
    "SetWorkspaceStack": 1,
    "Build": 1,
    "Project": 1,
}
analyze(folder, words)
```

	Word	Count	Search	Count	Markup	Ratio	\
0	Stack		85		1	85.0	
1	DisplayName		75		1	75.0	
2	Build		54		1	54.0	
3	Components		75		2	37.5	
4	Project		36		1	36.0	
5	Manifest		26		1	26.0	
6	CreateProject		5		1	5.0	
7	CreateWorkspace		7		2	3.5	
8	DestroyComponents		3		1	3.0	

9	UpdateResource	6	2	3.0
10	DestroyWorkspace	2	1	2.0
11	UpdateComponent	4	2	2.0
12	RestartWorkspaceComponents	1	1	1.0
13	StackByID	1	1	1.0
14	WatchJob	1	1	1.0
15	SetWorkspaceStack	1	1	1.0
16	RefreshCluster	1	1	1.0
17	RefreshResource	1	1	1.0
18	InitializeResource	1	1	1.0
19	ReconcileComponent	1	1	1.0
20	ForgetResource	1	1	1.0
21	TaskByID	1	1	1.0
22	DisposeResource	1	1	1.0
23	FinishTask	1	1	1.0
24	BuildWorkspace	1	1	1.0
25	DestroyStack	1	1	1.0
26	AcquireTask	1	1	1.0
27	FindEntity	1	1	1.0
28	FindManifest	1	1	1.0
29	ResourceByID	1	1	1.0
30	MakeManifest	1	1	1.0

	IsOutlier_MAD	ratio<30
0	True	False
1	True	False
2	True	False
3	True	False
4	True	False
5	True	True
6	True	True
7	True	True
8	True	True
9	True	True
10	True	True
11	True	True
12	False	True
13	False	True
14	False	True
15	False	True
16	False	True
17	False	True
18	False	True
19	False	True
20	False	True
21	False	True
22	False	True
23	False	True

24	False	True
25	False	True
26	False	True
27	False	True
28	False	True
29	False	True
30	False	True

	Word	Count	Search	Count	Markup	Ratio
0	MEAN	12.806452		1.129032	11.322581	

6.773809523809524

```
[167]: folder = 'e:\\phd\\test_repos_graphql2\\HydroAPI\\'
words = {
    "Merit": 1,
    "Strike": 1,
    "Giveaway": 1,
    "Server": 1,
    "User": 1,
    "Msgs": 1,
}
analyze(folder, words)
```

	Word	Count	Search	Count	Markup	Ratio	IsOutlier_MAD	ratio<30
0	Server		109		1	109.0	False	False
1	User		95		1	95.0	False	False
2	Giveaway		54		1	54.0	False	False
3	Merit		44		1	44.0	False	False
4	Strike		44		1	44.0	False	False
5	Msgs		7		1	7.0	False	True

	Word	Count	Search	Count	Markup	Ratio
0	MEAN	58.833333		1.0	58.833333	

47.666666666666664

```
[168]: folder = 'e:\\phd\\test_repos_graphql2\\mattermost-plugin-playbooks\\'
words = {
    "UpdateRunTaskActions": 1,
    "ChangeRunOwner": 1,
    "RemoveRunParticipants": 1,
    "AddRunParticipants": 1,
    "UpdateRun": 1,
    "SetRunFavorite": 1,
    "RemovePlaybookMember": 2,
    "AddPlaybookMember": 2,
    "DeleteMetric": 2,
    "UpdateMetric": 2,
    "AddMetric": 2,
```

```

    "UpdatePlaybook": 2,
    "UpdatePlaybookFavorite": 1,
    "Runs": 1,
    "Run": 1,
    "Playbooks": 1,
    "Playbook": 1
  }
analyze(folder, words)

```

	Word	Count	Search	Count	Markup	Ratio	IsOutlier_MAD	\
0	Run		463		1	463.0	True	
1	Playbook		353		1	353.0	True	
2	Playbooks		180		1	180.0	True	
3	Runs		55		1	55.0	True	
4	UpdateRunTaskActions		7		1	7.0	True	
5	UpdatePlaybook		12		2	6.0	True	
6	ChangeRunOwner		4		1	4.0	False	
7	UpdateRun		4		1	4.0	False	
8	UpdateMetric		7		2	3.5	False	
9	AddMetric		7		2	3.5	False	
10	DeleteMetric		7		2	3.5	False	
11	SetRunFavorite		3		1	3.0	False	
12	UpdatePlaybookFavorite		3		1	3.0	False	
13	AddRunParticipants		3		1	3.0	False	
14	RemoveRunParticipants		3		1	3.0	False	
15	RemovePlaybookMember		5		2	2.5	False	
16	AddPlaybookMember		5		2	2.5	False	

```

ratio<30
0    False
1    False
2    False
3    False
4    True
5    True
6    True
7    True
8    True
9    True
10   True
11   True
12   True
13   True
14   True
15   True
16   True

```

	Word	Count	Search	Count	Markup	Ratio
--	------	-------	--------	-------	--------	-------

```
0 MEAN      65.941176      1.352941  64.676471
6.570776255707763
```

```
[169]: folder = 'e:\\phd\\test_repos_graphql2\\SlabAPI\\'
words = {
    "GetViewCount": 2,
    "GetView": 1,
    "GetAllViews": 1,
    "GetUser": 1,
    "GetTrigger": 1,
    "GetAllTriggers": 1,
    "GetAllTickets": 1,
    "GetOrganization": 1,
}
analyze(folder, words)
```

	Word	Count	Search	Count	Markup	Ratio	IsOutlier_MAD	ratio<30
0	GetTrigger		11		1	11.0	False	True
1	GetOrganization		11		1	11.0	False	True
2	GetView		10		1	10.0	False	True
3	GetUser		10		1	10.0	False	True
4	GetAllTriggers		7		1	7.0	False	True
5	GetAllTickets		7		1	7.0	False	True
6	GetAllViews		6		1	6.0	False	True
7	GetViewCount		10		2	5.0	False	True

	Word	Count	Search	Count	Markup	Ratio
0	MEAN		9.0		1.125	8.375

27.11111111111111

```
[170]: folder = 'e:\\phd\\test_repos_graphql2\\sturdy\\'
words = {
    "Integrations": 1,
    "UpdatedGitHubPullRequestStatuses": 1,
    "PullCodebase": 1,
    "CreateOrUpdateCodebaseRemote": 1,
    "UpdateInstallation": 1,
    "DeleteIntegration": 1,
    "SetupGitHubRepository": 1,
    "Activity": 1,
    "DownloadZip": 1,
    "Changes": 1,
    "Members": 2,
    "LastUsedView": 2,
    "UpdatedOrganization": 1,
    "UpdatedSuggestion": 1,
    "UpdatedView": 1,
```



```

    "UpdatedWorkspaceDiffs": 1,
    "GenerateKeyPair": 1,
    "RemovePatches": 2,
    "DismissSuggestionHunks": 1,
    "AddPublicKey": 1,
    "WatchWorkspace": 1,
    "RequestReview": 1,
    "UpdateACL": 1,
    "ArchiveNotifications": 1,
    "AddUserToCodebase": 1,
    "VerifyEmail": 2,
    "UpdateUser": 2,
    "DeleteComment": 1,
    "LandWorkspaceChange": 2,
    "Comment": 3,
    "Workspaces": 2,
    "ResolvedBy": 1,
    "Notifications": 1,
    "UpdateComment": 1,
    "CreateComment": 1,
    "UpdateCodebase": 1,
    "ReportWorkspacePresence": 1,
    "DismissSuggestion": 1,
    "UpdateOrganization": 1,
    "UpdatedWorkspacesStatuses": 1,
    "CreateOrUpdateGitHubPullRequest": 1,
    "TriggerInstantIntegration": 1,
    "PushCodebase": 1,
    "UpdatedGitHubPullRequest": 1,
}
analyze(folder, words)

```

	Word	Count	Search	Count	Markup	Ratio	\
0	Activity		43		1	43.000000	
1	Comment		94		3	31.333333	
2	LandWorkspaceChange		22		2	11.000000	
3	Members		19		2	9.500000	
4	AddPublicKey		7		1	7.000000	
5	ResolvedBy		6		1	6.000000	
6	Changes		6		1	6.000000	
7	UpdateInstallation		4		1	4.000000	
8	DownloadZip		4		1	4.000000	
9	CreateOrUpdateGitHubPullRequest		4		1	4.000000	
10	CreateComment		4		1	4.000000	
11	RemovePatches		7		2	3.500000	
12	VerifyEmail		7		2	3.500000	
13	UpdatedGitHubPullRequestStatuses		3		1	3.000000	
14	UpdatedGitHubPullRequest		3		1	3.000000	

15	SetupGitHubRepository	3	1	3.000000
16	CreateOrUpdateCodebaseRemote	3	1	3.000000
17	ReportWorkspacePresence	2	1	2.000000
18	UpdateCodebase	2	1	2.000000
19	DismissSuggestion	2	1	2.000000
20	UpdatedWorkspacesStatuses	2	1	2.000000
21	UpdateComment	2	1	2.000000
22	Notifications	2	1	2.000000
23	TriggerInstantIntegration	2	1	2.000000
24	Workspaces	4	2	2.000000
25	PushCodebase	2	1	2.000000
26	UpdateOrganization	2	1	2.000000
27	Integrations	2	1	2.000000
28	DeleteComment	2	1	2.000000
29	UpdateUser	4	2	2.000000
30	AddUserToCodebase	2	1	2.000000
31	ArchiveNotifications	2	1	2.000000
32	RequestReview	2	1	2.000000
33	WatchWorkspace	2	1	2.000000
34	DismissSuggestionHunks	2	1	2.000000
35	GenerateKeyPair	2	1	2.000000
36	UpdatedWorkspaceDiffs	2	1	2.000000
37	UpdatedView	2	1	2.000000
38	UpdatedSuggestion	2	1	2.000000
39	UpdatedOrganization	2	1	2.000000
40	LastUsedView	4	2	2.000000
41	DeleteIntegration	2	1	2.000000
42	PullCodebase	2	1	2.000000
43	UpdateACL	2	1	2.000000

	IsOutlier_MAD	ratio<30
0	True	False
1	True	False
2	True	True
3	True	True
4	True	True
5	True	True
6	True	True
7	True	True
8	True	True
9	True	True
10	True	True
11	True	True
12	True	True
13	True	True
14	True	True
15	True	True
16	True	True

17	False	True
18	False	True
19	False	True
20	False	True
21	False	True
22	False	True
23	False	True
24	False	True
25	False	True
26	False	True
27	False	True
28	False	True
29	False	True
30	False	True
31	False	True
32	False	True
33	False	True
34	False	True
35	False	True
36	False	True
37	False	True
38	False	True
39	False	True
40	False	True
41	False	True
42	False	True
43	False	True

	Word	Count	Search	Count	Markup	Ratio
0	MEAN		6.795455		1.204545	4.609848
						10.017857142857142

```
[171]: folder = 'e:\\phd\\test_repos_graphql2\\toolkit-for-cardano\\'
words = {
    "WalletDelegate": 1,
    "WalletRegister": 1,
    "WalletFund": 1,
    "WalletCreate": 1,
    "SendFunds": 1,
    "TxSubmit": 1,
    "TxSign": 1,
    "TxBuild": 1,
    "Mint": 1,
    "Wallets": 1,
    "Utxos": 2,
    "TxFee": 1,
}
```

```
analyze(folder, words)
```

	Word	Count	Search	Count	Markup	Ratio	IsOutlier_MAD	ratio<30
0	Utxos		22		2	11.0	True	True
1	Mint		9		1	9.0	True	True
2	WalletDelegate		2		1	2.0	False	True
3	WalletRegister		2		1	2.0	False	True
4	TxSubmit		2		1	2.0	False	True
5	TxSign		2		1	2.0	False	True
6	TxFee		2		1	2.0	False	True
7	WalletFund		1		1	1.0	False	True
8	WalletCreate		1		1	1.0	False	True
9	SendFunds		1		1	1.0	False	True
10	TxBuild		1		1	1.0	False	True
11	Wallets		1		1	1.0	False	True

	Word	Count	Search	Count	Markup	Ratio
0	MEAN	3.833333	1.083333	2.916667		

12.969696969696969

```
[ ]:
```

```
[ ]:
```

```
[ ]:
```

```
[135]: data = global_array
#
bins = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 30, 50, 100, 7000]
```

```
#
intervals = [
    '1', '2', '3', '4', '5', '6', '7', '8', '9', '10',
    '11-15', '16-20', '21-30', '31-50', '51-100', '>100'
]
```

```
#
counts, _ = np.histogram(data, bins=bins)
```

```
# { : }
result = dict(zip(intervals, counts))
print(result)
```

```
{'1': 56, '2': 50, '3': 22, '4': 20, '5': 7, '6': 7, '7': 13, '8': 13, '9': 11,
'10': 24, '11-15': 8, '16-20': 12, '21-30': 17, '31-50': 22, '51-100': 43,
'>100': 0}
```

```
[150]: print(len(data[data==1]))
print(len(data[(data>1)&(data<=4)]))
print(len(data[(data>5)&(data<=10)]))
print(len(data[(data>10)]))
```

56
91
51
119

```
[109]: # folder = 'e:\\phd\\test_repos\\modules\\'
# words = {
#     "CompleteFollowup": 1,
#     "ReAssignFollowup": 2,
#     "RemovePharmacy": 1,
#     "SetABVariant": 1,
#     "BlockPatient": 1,
#     "SetMediaRead": 1,
#     "ValidateAppointmentInsurance": 3,
#     "UpdateProviderInsurance": 1,
#     "EditStateCompanies": 1,
#     "UpdateEncounterScreenedStatus": 1,
#     "StartPaymentTransaction": 1,
#     "SubscribeToPhoneCallEvents": 1,
#     "SetAppointmentType": 1,
#     "MakeCardDefault": 2,
#     "DeletePatientNote": 1,
#     "UpdatePatientNote": 1,
#     "SetVisitTypeOrders": 2,
#     "SetVisitTypeVisibility": 2,
#     "AddICD10MedicalCode": 2,
#     "AddMedicalCodes": 1,
#     "InviteClient": 1,
#     "DeleteEvent": 1,
#     "NeedFollowup": 1,
#     "TwilioInfo": 1,
#     "CurrentTime": 1,
#     "Person": 1,
#     "PHIData": 1,
#     "Clients": 1,
#     "Messages": 1,
#     "PastVisits": 1,
#     "Appointment": 1,
#     "CheckInData": 1,
#     "NoteTemplate": 1,
#     "VisitType": 1,
#     "ScheduleVisit": 1,
#     "UpdateAvailability": 1,
```

```
#      "AssignQuestionnaire": 1,
#      "PublishForm": 1,
#      "LoadFollowupByPatients": 1,
#      "InsuranceCosts": 1,
#      "PatientDetails": 1,
#      "ConsentForms": 1
# }
# analyze(folder, words)
```

```
[95]: # folder = 'e:\\phd\\test_repos_graphql2\\bramble\\'
# words = {
#      "UploadGadgetFile": 1,
#      "UploadGizmoFile": 1,
#      "Gizmo": 2,
#      "Gizmos": 2,
#      "Delay": 1,
#      "Gizmos": 2,
#      "RandomGizmo": 1,
# }
# analyze(folder, words)
```

```
[110]: # folder = 'e:\\phd\\test_repos_graphql2\\go-graphql-starter\\'
# words = {
#      "Signup": 1,
#      "User": 1,
#      "Login": 1,
#      "Viewer": 1,
# }
# analyze(folder, words)
```

```
[105]: # folder = 'e:\\phd\\test_repos\\backend\\fsm-gateway\\'
# words = {
#      "GetSession": 1,
#      "GetDocumentData": 1,
#      "GetDocumentText": 1,
#      "GetConsentList": 1,
#      "StartSession": 1,
#      "SendDocumentData": 1,
#      "FinishSession": 1,
#      "StartAuthSessionByPhone": 1,
#      "GetJwtByPhoneCode": 1,
#      "GetJwtByRefreshToken": 1
# }
# df = analyze(folder, words)
```

```
[106]: # folder = 'e:\\phd\\test_repos\\LP\\delivery-offering\\'
# words = {
```

```

#     "DeliveryTimeslots": 2,
#     "DeliveryTimeslot": 2,
#     "DeliveryTypes": 2,
#     "ReserveDeliveryTimeslot": 2,
#     "LookupDeliveryOffers": 2,
#     "DeliveryOfferByID": 2,
#     "AcceptDeliveryOffer": 2,
#     "ValidateDeliveryOffer": 2,
# }
# analyze(folder, words)

```

```

[107]: # folder = 'e:\\phd\\test_repos\\LP\\warehouses\\'
# words = {
#     "TerminateWarehouses": 2,
#     "UpdateWarehouses": 2,
#     "CreateWarehouses": 2,
#     "Warehouses": 2,
# }
# analyze(folder, words)

```

```

[108]: # folder = 'e:\\phd\\test_repos\\LP\\channel-profile\\'
# words = {
#     "CreateChannelProfile": 2,
#     "UpdateChannelProfile": 2,
#     "DeleteChannelProfile": 2,
#     "ChannelProfiles": 2,
#     "ChannelsByFilter": 2,
# }
# analyze(folder, words)

```